

Development of a Deep Reinforcement Learning Agent for Optimal Project Portfolio Budgeting under Uncertain Cash Flows

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Abstract

Project portfolio budget allocation for EPC contractors in the oil and gas sector is a sequential decision-making problem under uncertain cash inflows, cash outflows, and project performance. Each portfolio consists of multiple projects with heterogeneous attributes, including budget at completion, start and finish dates, contractual S-curves, fixed profit margins, milestone-based payment structures, and uncertain contractor performance represented through Earned Value Management metrics such as cumulative SPI and CPI. The model also captures delayed and uncertain milestone payments, working-capital shortages, credit usage during negative-cashflow periods, and management intervention actions that affect cashflow and performance trajectories.

Despite its practical importance, rigorous mathematical modeling of project portfolio budgeting under joint cashflow and performance uncertainty remains largely absent in recent literature, while most AI/ML-related studies discuss the topic qualitatively. This paper proposes a custom stochastic reinforcement learning environment for learning adaptive budget allocation policies under unknown future uncertainty. A PPO-based agent is trained with a multi-objective reward function to allocate constrained budgets efficiently, prioritize economically viable projects, avoid loss-making allocations, and learn optimal allocation timing. The framework provides a foundational RL model for EPC oil and gas portfolio budgeting, supports future fine-tuning with anonymized industry data, and enables immediate next-period allocation recommendations as well as rolling 12-period trajectory simulation.

Keywords: Project Portfolio Budgeting, Reinforcement Learning, EPC Contractors, Oil and Gas Projects, Cashflow Uncertainty, Earned Value Management, PPO, Rolling Horizon Control

1 Problem Formulation

2 Problem Formulation

2.1 Overview

This section develops the formal problem structure for portfolio-level budget allocation in contracted project portfolios. The formulation proceeds in three levels of increasing

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generality, each building directly on the preceding one.

Level 1 — Deterministic Full-Foresight Linear Program. All contractor performance parameters, expenditure trajectories, and payment timing are treated as known with certainty. The resulting program is a pure linear program (LP) in the continuous budget allocation variables. Because no real decision-maker possesses this information *ex ante*, the Level 1 optimum constitutes a theoretical upper bound on portfolio performance: no implementable policy can exceed it. This upper bound anchors the empirical comparison in Section ??.

Level 2 — Stochastic Extension and Intractability. Deterministic performance parameters are replaced by stochastic processes consistent with the empirical EVM literature. The resulting multi-stage stochastic program is formally characterized, and the combinatorial growth of its scenario tree with the number of projects and planning periods is established. This intractability argument motivates the reformulation in Level 3.

Level 3 — Constrained Markov Decision Process. The stochastic program of Level 2 is reformulated as a Constrained Markov Decision Process (CMDP). An explicit mapping is provided from LP elements to MDP primitives: decision variables map to actions, constraint parameters map to state variables, the objective maps to the reward function, and feasibility constraints define the constrained action space. The CMDP is implemented as a configurable, Gymnasium-compatible synthetic environment; a Proximal Policy Optimization (PPO) agent trained on this environment serves as the primary computational demonstration of the framework’s utility.

The three levels are not alternative models of the same problem. They are a coherent modeling arc: Level 1 defines the problem’s theoretical ceiling, Level 2 characterizes why that ceiling is computationally unreachable, and Level 3 provides the tractable reformulation that makes the problem solvable at realistic scale. The remainder of this section develops each level in turn, preceded by the shared problem primitives on which all three levels draw.

The formulation integrates domain knowledge from project management and operations research: S-curve expenditure dynamics ??, Earned Value Management (EVM) performance tracking ?, milestone-based payment structures consistent with FIDIC contract forms ???, and the theory of investment under uncertainty as it applies to abandonment options ?. The LP of Level 1 is situated within the stochastic programming tradition ?; the CMDP of Level 3 is situated within the constrained MDP literature ?, framed as a direct descendant of Bellman’s dynamic programming and stochastic control theory.

2.2 Scope and Positioning

The problem addressed here is operational, not strategic. The literature on project portfolio management has devoted considerable attention to the *selection* problem: given a candidate set of projects and a capital budget, which subset should the organization commit to ??? This work takes the selection decision as given. A portfolio of contracted projects exists; commitments have been signed; and the relevant decision-maker — a portfolio manager or governing board — faces a recurring operational question: how should the available budget be allocated across active projects at each discrete planning period?

This is the budget allocation control problem, and it is the sole focus of the formulation. The decision-maker exercises one lever: the amount of budget directed to each active project at each period. All strategic and operational outcomes of interest — milestone achievement, cash flow balance, schedule adherence, and ultimate project success or termination — flow through this single allocation decision. The formulation makes this causal chain explicit and embeds it within a sequential decision-making framework.

The model is deliberately general. Every structural element — the number of projects, the milestone structure, the S-curve shape parameters, the advance payment and retention ratios, the performance response distribution, the deviation penalty weight, and the termination settlement mechanics — is parameterized rather than fixed. Sector-specific instantiation is achieved through empirical calibration of these parameters, not through changes to the model structure. The calibration used in this work draws on anonymized data from two major EPC contractors, supplemented by World Bank ICR data ? and infrastructure cost overrun datasets ?. The general structure is applicable across construction, software development, defense contracting, and other project-intensive sectors ??.

2.3 Assumptions

Four structural choices underlie the entire formulation. Each is stated here with its justification, because they determine the form of every constraint and variable that follows.

A portfolio is an aggregation of independent bilateral contracts. Each project in the portfolio corresponds to a separate contract between the client and a distinct contractor. Projects do not share resources, workforces, or contractual obligations within the formulation; interaction occurs only through the shared portfolio budget constraint. This independence structure is consistent with standard EPC contracting practice and permits the portfolio problem to be decomposed into project-level dynamics aggregated at the portfolio level. It is also the structure that makes the LP of Level 1 tractable: the coupling across projects enters only through the single budget constraint at each period.

The resource allocation problem reduces to budget allocation. Within each project, the contractor controls workforce, materials, equipment, and subcontracting. From the portfolio manager’s perspective, the only instrument available is the budget directed to each project. The contractor’s internal response to that budget — the efficiency with which it is converted into physical progress — is modeled as a performance parameter, stochastic at Level 2 and treated as known at Level 1. This reduction of the resource allocation problem to a scalar budget allocation per project per period is both a modeling choice and a reflection of the actual decision authority of a portfolio-level governing body.

Time is discrete, with periods representing calendar months. The planning horizon $\mathcal{T} = \{1, \dots, H\}$ is indexed by discrete periods. Monthly periods are adopted because the key governance events in portfolio management — progress reporting, payment certification, performance review meetings — are conventionally held on a monthly cycle. The LP of Level 1 optimizes over the full horizon simultaneously; the CMDP of Level 3 operates period by period, with decisions made sequentially as new information is revealed.

Contract mechanics are calibrated to FIDIC forms. The milestone-based payment structure, advance payment and recovery mechanism, retention and release, and termination settlement mechanics are all specified to be consistent with FIDIC Red, Yellow, and Silver Book contract forms ?. This anchors the formulation in the dominant international EPC contracting framework and ensures that the model parameters have direct empirical counterparts in contractor financial data. Where FIDIC allows variation by agreement

(e.g., the advance payment ratio, the retention percentage, the deviation penalty weight), these are treated as project-specific calibrated parameters rather than fixed constants.

2.4 Problem Primitives

This subsection establishes the complete notation, project-level parameters, state variables, transition dynamics, and portfolio aggregation structure on which all three formulation levels draw. The organization mirrors the environment's ontology: objects are defined first (projects and their parameters), then their state, then the dynamics that move state forward, then the portfolio composition, and finally the shared objective. Nothing introduced here is level-specific; every symbol defined below appears without redefinition across Level 1, Level 2, and Level 3.

2.4.1 Sets and Indices

All symbols introduced in subsequent subsections are indexed over the sets defined here. This subsection is exhaustive: every set, index, and structured collection that appears anywhere in the formulation is declared below.

Primary index sets.

$\mathcal{P} = \{1, \dots, n\}$ Set of projects in the portfolio, indexed by i . All projects are committed at the start of the planning horizon; no new projects enter mid-horizon.

$\mathcal{T} = \{1, \dots, H\}$ Set of discrete planning periods, indexed by t , each representing one calendar month.

$\mathcal{M}_i = \{1, \dots, M_i\}$ Ordered set of contractual milestones for project i , indexed by j . Milestones are ordered by their cumulative progress thresholds: $\theta_{i,1} < \theta_{i,2} < \dots < \theta_{i,M_i} = 1$.

Derived and dynamic sets.

$\mathcal{A}_i^{\text{plan}} = \{t : s_i \leq t \leq f_i\}$ Planned execution window of project i , spanning its planned start s_i through its planned finish f_i . All S-curve and deviation computations are anchored to this window.

$\mathcal{P}_t^{\text{active}} \subseteq \mathcal{P}$ Subset of projects active at period t . In Level 1, this set is determined ex ante from the known efficiency trajectories $\{\bar{\eta}_{i,t}\}$ and is a fixed parameter. In Level 2 and Level 3, it is endogenous, evolving with realized status transitions.

$\mathcal{M}_i^{\text{cert}}(t) \subseteq \mathcal{M}_i$ Subset of milestones for project i certified up to and including period t :

$$\mathcal{M}_i^{\text{cert}}(t) = \{j \in \mathcal{M}_i : \theta_{i,j} \leq P_{i,t}^{\text{actual}}\}.$$

This set grows monotonically over time and governs the release of milestone payments and advance recovery deductions.

$\mathcal{P}^{\text{comp}} \subseteq \mathcal{P}$, $\mathcal{P}^{\text{term}} \subseteq \mathcal{P}$ Subsets of projects that have reached **completed** or **terminated** status, respectively, by the end of the planning horizon. These partition $\mathcal{P}$ together with $\mathcal{P}_H^{\text{active}}$.

Decision variable.

$x_{i,t} \geq 0$ Budget allocated to project $i \in \mathcal{P}$ at period $t \in \mathcal{T}$. This is the sole decision variable of the portfolio manager. The action at period t is the vector $\mathbf{x}_t = (x_{i,t})_{i \in \mathcal{P}_t^{\text{active}}}$.

Portfolio-level scalar.

$B_t \geq 0$ Available portfolio cash balance at the beginning of period t . B_1 is given as an initial condition. The balance evolves through the portfolio aggregation dynamics defined in Section ??.

2.4.2 Project Parameters

Each project $i \in \mathcal{P}$ is fully characterized at commitment by the contractual and operational parameters in Tables ?? and ??. These parameters are fixed at the start of the planning horizon and do not change during execution. Parameters are grouped by functional role: schedule and contract structure, then cost, penalty, and termination mechanics.

Parameter	Symbol	Domain	Definition
Planned start & finish Schedule	s_i, f_i	$\mathbb{Z}^+$	Planned start and finish periods for project i . Anchor the execution window $\mathcal{A}_i^{\text{plan}}$ and all S-curve computations.
Planned duration Schedule	D_i^{plan}	$\mathbb{Z}^+$	$D_i^{\text{plan}} = f_i - s_i + 1$. Baseline for normalizing actual progress against plan.
S-curve shape parameters Schedule	a_i, b_i	$\mathbb{R}^+$	Shape parameters of the regularized incomplete Beta function $F_i(\tau) = I_\tau(a_i, b_i)$ governing the planned expenditure profile. Calibrated empirically per project.
Milestone threshold Schedule	$\theta_{i,j}$	$(0, 1]$	Cumulative progress fraction required to certify milestone j of project i . Milestones are ordered: $\theta_{i,1} < \dots < \theta_{i,M_i} = 1$.
Allowable delay buffer Schedule	buffer_i	$\mathbb{Z}_0^+$	Number of periods beyond f_i during which project i remains active before forced termination is assessed.
Contract price Contract	CP_i	$\mathbb{R}^+$	Total payment from client: $CP_i = \text{BAC}_i(1 + \pi_i)$, where π_i is the contractual profit margin.
Profit margin Contract	π_i	$\mathbb{R}_0^+$	Contractual markup on BAC_i ; determines the spread between contractor cost and client-facing price.
Advance payment ratio Contract	α_i	$[0, 1]$	Fraction of CP_i disbursed at project initiation; $A_i = \alpha_i \cdot CP_i$. Recovered progressively through milestone deductions.
Retention ratio Contract	ρ_i	$[0, 1]$	Fraction withheld from each milestone payment; released in full upon project completion as $R_i^{\text{ret}} = \rho_i \cdot CP_i$.
Milestone payment weight Contract	$\phi_{i,j}$	$\mathbb{R}^+$	Fraction of CP_i released at milestone j ; weights satisfy $\sum_{j \in \mathcal{M}_i} \phi_{i,j} = 1$.

Table 1: Schedule and contract parameters for project $i \in \mathcal{P}$. All parameters are fixed at commitment.

Parameter	Symbol	Domain	Definition
Budget at completion Cost	BAC_i	$\mathbb{R}^+$	Total planned expenditure from the contractor's perspective; the planned cost basis against which EVM indices are computed.
Discount factor Cost	γ	$(0, 1]$	Per-period discount factor applied to future cash flows in the objective. Shared across all projects.
Initial cash balance Cost	B_1	$\mathbb{R}^+$	Portfolio cash balance at the start of the planning horizon. Initial condition for the balance recursion.
Deviation penalty weight Penalty	λ_i	$\mathbb{R}_0^+$	Per-period cost per unit of absolute progress deviation from plan. Calibrated so that the cumulative penalty at the planned finish date, under a sustained delay equal to the contractual limit, equals $\Lambda_i \cdot CP_i$.
Deviation penalty cap Penalty	Λ_i	$\mathbb{R}_0^+$	Maximum cumulative deviation penalty as a fraction of CP_i . Applies symmetrically to schedule underrun and overrun; mirrors the contractual delay penalty cap.
Termination threshold Termination	κ_i	$(0, 1]$	Fraction of the penalty cap $\Lambda_i \cdot CP_i$ at which the project becomes a candidate for termination. Signals exhaustion of schedule tolerance.
Minimum termination allocation Termination	m_i^0	$\mathbb{R}_0^+$	Minimum budget allocation required to hold a project active and service its contractual obligations. Falling to this floor for k consecutive periods, in conjunction with penalty saturation, triggers the termination assessment and initiates the settlement defined in Section ??.
Termination confirmation window Termination	k	$\mathbb{Z}^+$	Number of consecutive periods at the minimum allocation floor required to confirm an abandonment decision and trigger termination.

Table 2: Cost, penalty, and termination parameters for project $i \in \mathcal{P}$. All parameters are fixed at commitment.

Planned Expenditure Profile. The planned cumulative expenditure of project i follows a normalized S-curve ??, parameterized by (a_i, b_i) from Table ??. Define the normalized planned time at period t as:

$$\tau_{i,t} = \frac{t - s_i + 1}{D_i^{\text{plan}}}, \quad t \in \mathcal{A}_i^{\text{plan}},$$

with the clamped value $\tau_{i,t}^c = \min\{1, \max\{0, \tau_{i,t}\}\}$ handling periods outside the planned window. The planned cumulative expenditure fraction and its period increment are:

$$F_i(\tau) = I_\tau(a_i, b_i), \quad \bar{E}_{i,t} = BAC_i \cdot F_i(\tau_{i,t}^c), \quad \Delta \bar{E}_{i,t} = \bar{E}_{i,t} - \bar{E}_{i,t-1},$$

where $I_\tau(\cdot, \cdot)$ is the regularized incomplete Beta function. Under the nominal plan, planned cumulative progress tracks planned expenditure fraction:

$$P_{i,t}^{\text{plan}} = F_i(\tau_{i,t}^c).$$

This profile is a fixed project parameter derived at commitment from $(a_i, b_i, s_i, f_i, \text{BAC}_i)$. It is not a decision variable; it serves as the baseline against which actual execution is measured in every deviation and EVM computation that follows.

2.4.3 State Variables

State variables describe the condition of each project and the portfolio at the end of each period. They are grouped into two tiers that reflect the environment's ontology: project-level states capture the execution history and performance of individual projects; the portfolio-level state aggregates these into the quantities relevant to the allocation decision.

Project-level state. For each $i \in \mathcal{P}$ and $t \in \mathcal{T}$:

$P_{i,t}^{\text{actual}} \in [0, 1]$ *Cumulative actual progress : the fraction of project scope physically completed at the end of period t .*

$E_{i,t} \geq 0$ Cumulative actual expenditure on project i up to and including period t . Initialized at $E_{i,0} = 0$.

$\text{SPI}_{i,t}$ Schedule Performance Index:

$$\text{SPI}_{i,t} = \frac{P_{i,t}^{\text{actual}}}{P_{i,t}^{\text{plan}}}, \quad P_{i,t}^{\text{plan}} > 0.$$

Values below unity indicate schedule underperformance; values above unity indicate acceleration ahead of plan.

$\text{CPI}_{i,t}$ Cost Performance Index:

$$\text{CPI}_{i,t} = \frac{P_{i,t}^{\text{actual}} \cdot \text{BAC}_i}{E_{i,t}}, \quad E_{i,t} > 0.$$

Values below unity indicate cost overrun relative to work performed.

$\delta_{i,t}$ Signed progress deviation:

$$\delta_{i,t} = P_{i,t}^{\text{actual}} - P_{i,t}^{\text{plan}}.$$

Negative values indicate schedule lag; positive values indicate acceleration beyond the planned profile.

$\Delta_{i,t}^{\text{pen}} \geq 0$ Cumulative deviation penalty accrued up to period t . Defined and computed in Section ??.

$\text{status}_{i,t} \in \{\text{active}, \text{completed}, \text{terminated}\}$ Categorical project status at the end of period t . Transitions are governed by the progress recursion and the termination conditions defined in Sections ?? and ??.

Portfolio-level state. The portfolio state at period t is the composition of all project states together with the shared cash balance:

$$S_t = \left(\{P_{i,t}^{\text{actual}}, E_{i,t}, \text{SPI}_{i,t}, \text{CPI}_{i,t}, \delta_{i,t}, \Delta_{i,t}^{\text{pen}}, \text{status}_{i,t}\}_{i \in \mathcal{P}}, B_t \right).$$

The dimension of S_t scales linearly with n . The cash balance B_t is the sole portfolio-level quantity that couples project dynamics: it is the shared resource over which allocation decisions are contested at each period, and its evolution is governed by the aggregation defined in Section ??.

2.4.4 Transition Dynamics

The transition dynamics determine how the portfolio state S_t evolves to S_{t+1} given the allocation decision $\mathbf{x}_t$. Three mechanisms govern this transition: the conversion of allocated budget into physical progress, the contractual cash-flow mechanics that translate progress into payments, and the running deviation penalty that prices schedule adherence.

Budget-to-Progress Conversion. Allocated budget $x_{i,t}$ is converted to a progress increment through an efficiency parameter:

$$\Delta P_{i,t}^{\text{actual}} = \eta_{i,t} \cdot \frac{x_{i,t}}{\text{BAC}_i},$$

where $\eta_{i,t}$ is the *budget-to-progress efficiency* for project i at period t . This scalar captures all sources of conversion uncertainty — workforce productivity, supply chain conditions, technical difficulty, and weather — that determine how much physical progress a given budget expenditure yields in practice.

Treatment across levels. In Level 1, $\eta_{i,t}$ is replaced by a known deterministic value $\bar{\eta}_{i,t}$ calibrated from the project’s planned performance profile; under the nominal plan, $\bar{\eta}_{i,t} = 1$ implies that expenditure tracks the S-curve exactly. In Level 2 and Level 3, $\eta_{i,t}$ is a random primitive realized only after $x_{i,t}$ is committed, with a distribution calibrated from empirical EVM data.

The project-level state recursion follows directly:

$$P_{i,t}^{\text{actual}} = P_{i,t-1}^{\text{actual}} + \eta_{i,t} \cdot \frac{x_{i,t}}{\text{BAC}_i}, \quad E_{i,t} = E_{i,t-1} + x_{i,t},$$

with $P_{i,0}^{\text{actual}} = 0$ and $E_{i,0} = 0$.

Contractual Cash-Flow Mechanics. Project cash flows consist of outflows (budget allocations $x_{i,t}$) and inflows (contractual payments). The payment structure is consistent with FIDIC contract forms ? and comprises three mechanisms.

Advance Payment. At project initiation (period s_i), the client disburses:

$$A_i = \alpha_i \cdot \text{CPI}_i.$$

The advance payment is a contractual timing adjustment, not a financing instrument: it carries no interest obligation and is recovered progressively through deductions from subsequent milestone payments.

Advance Recovery. The recovery deduction applied at milestone j is proportional to the milestone payment weight:

$$\Delta A_{i,j} = \phi_{i,j} \cdot A_i.$$

This ensures the full advance is recovered by the time all milestones are certified, regardless of project duration.

Milestone Certification and Net Payment. Milestone $j \in \mathcal{M}_i$ is certified at the first period t such that $P_{i,t}^{\text{actual}} \geq \theta_{i,j}$. On certification, the gross milestone payment, net payment after retention, and net payment after both retention and advance recovery are:

$$R_{i,j} = \phi_{i,j} \cdot CP_i \cdot (1 - \rho_i), \quad R_{i,j}^{\text{net}} = R_{i,j} - \Delta A_{i,j}.$$

Retention and advance recovery operate independently: recovery deductions are applied to gross payments prior to retention. The accounting identity across all payment flows is:

$$A_i + \sum_{j \in \mathcal{M}_i} R_{i,j}^{\text{net}} + R_i^{\text{ret}} = CP_i,$$

where $R_i^{\text{ret}} = \rho_i \cdot CP_i$ is the total retention accumulated across milestones and released at completion.

Final Payment and Retention Release. Upon completion ($P_{i,t}^{\text{actual}} = 1$), the final net inflow is:

$$R_i^{\text{final}} = R_{i,M_i}^{\text{net}} + R_i^{\text{ret}}.$$

No separate lump-sum delay penalty appears here; schedule adherence is enforced entirely through the continuous running deviation penalty defined next. The retention release and final milestone payment together constitute the natural completion incentive embedded in the payment structure.

Symmetric Running Deviation Penalty. Schedule adherence is priced continuously through a symmetric running penalty assessed at every period $t \in \mathcal{T}$. The per-period penalty for project i is:

$$c_{i,t}^{\text{dev}} = \min\{\lambda_i \cdot |\delta_{i,t}|, \Lambda_i \cdot CP_i\},$$

where $\delta_{i,t} = P_{i,t}^{\text{actual}} - P_{i,t}^{\text{plan}}$ is the signed progress deviation, $\lambda_i \geq 0$ is the per-unit penalty weight, and Λ_i is the cap as a fraction of CP_i .

Three design properties of this specification are explicit.

Symmetry. The penalty applies to $|\delta_{i,t}|$, not to $\max\{0, -\delta_{i,t}\}$. Both schedule lag and acceleration are penalized. This discourages the decision-maker from pushing projects ahead of plan, which consumes budget inefficiently and risks scope or quality degradation ?.

Continuity. The penalty is assessed every period, not only at contractual completion. This provides a continuously updating signal that feeds the termination assessment in Section ??: a project whose cumulative penalty grows monotonically signals persistent underperformance before any contractual deadline is breached.

Calibration to contractual terms. The weight λ_i is calibrated so that the cumulative penalty at the planned finish date, under a sustained delay equal to the contractual limit, equals $\Lambda_i \cdot CP_i$. This connects the running signal to the actual contractual delay-penalty exposure assigned by FIDIC terms ?.

The cumulative deviation penalty up to period t is:

$$\Delta_{i,t}^{\text{pen}} = \sum_{\tau=s_i}^t c_{i,\tau}^{\text{dev}}.$$

This quantity enters both the objective function as a running cost and the termination condition as a trigger threshold.

2.4.5 Termination Conditions and Settlement

A project transitions from **active** to **terminated** at the first period t at which both of the following conditions hold simultaneously:

1. **Penalty saturation.** The cumulative deviation penalty has reached or exceeded the contractual ceiling:

$$\Delta_{i,t}^{\text{pen}} \geq \kappa_i \cdot \Lambda_i \cdot CP_i.$$

This indicates that the project has exhausted its schedule tolerance and the client has lost contractual leverage over the contractor.

2. **Confirmed allocation floor.** The portfolio manager has allocated no more than the minimum termination allocation m_i^0 for k consecutive periods:

$$x_{i,\tilde{t}} \leq m_i^0 \quad \forall \tilde{t} \in \{t - k + 1, \dots, t\}.$$

Holding allocation at m_i^0 is the mechanism by which the portfolio manager signals economic abandonment: it is the lowest level at which contractual obligations are serviced without active investment. Sustaining this floor for k periods confirms that the decision to cease investment is deliberate rather than transient.

The role of m_i^0 is therefore specific and bounded: it is not a performance-recovery instrument, but the floor allocation required to hold a project in a legally serviceable state while the termination conditions are evaluated. It has no other role in the formulation.

The simultaneous satisfaction of both conditions ensures termination occurs only when the project is both contractually disadvantageous (condition 1) and operationally abandoned (condition 2).

Termination Settlement. Upon termination at period t_i^{term} , the net settlement payment is:

$$\begin{aligned} V_i^{\text{term}} &= P_{i,t_i^{\text{term}}}^{\text{actual}} \cdot CP_i, \\ A_i^{\text{outstanding}} &= A_i - \sum_{j \in \mathcal{M}_i^{\text{cert}}(t_i^{\text{term}})} \Delta A_{i,j}, \\ R_i^{\text{term}} &= V_i^{\text{term}}(1 - \rho_i) - A_i^{\text{outstanding}}. \end{aligned}$$

If $R_i^{\text{term}} < 0$, the outstanding advance payment balance constitutes a net obligation settled from available portfolio cash inflows ahead of other commitments, consistent with FIDIC termination clauses ?.

2.4.6 Portfolio Aggregation

The portfolio $\mathcal{P}$ aggregates n projects whose individual dynamics are coupled through the shared cash balance B_t . The cash balance evolves as:

$$B_{t+1} = B_t - \sum_{i \in \mathcal{P}_t^{\text{active}}} x_{i,t} + V_t, \quad \forall t \in \mathcal{T},$$

where the aggregate inflow V_t collects all contractual payments received at period t :

$$V_t = \sum_{i \in \mathcal{P}} \sum_{j \in \mathcal{M}_i} R_{i,j}^{\text{net}} \cdot \mathbf{1}\{\text{milestone } (i,j) \text{ newly certified at } t\} + \sum_{i \in \mathcal{P}_t^{\text{term}}} R_i^{\text{term}},$$

with $\mathcal{P}_t^{\text{term}}$ the set of projects newly terminated at period t . The constraint $B_t \geq 0$ is enforced at all periods; no borrowing or credit facility is available. This hard constraint eliminates the soft budget dynamics present in formulations that allow internal credit and is consistent with the governance structure of a portfolio-level board operating under a fixed capital commitment ?.

The portfolio state S_t defined in Section ?? therefore evolves as a deterministic function of $\mathbf{x}_t$ under Level 1, and as a controlled stochastic process under Level 2 and Level 3. The linear scaling of $\dim(S_t)$ with n , combined with the full-horizon coupling through B_t , characterizes the computational complexity of the portfolio allocation problem and motivates the intractability argument in Level 2.

2.4.7 Objective

The decision-maker seeks to maximize the discounted cumulative net cash flow over the planning horizon:

$$\max_{\{x_{i,t}\}} \mathbb{E} \left[\sum_{t=1}^H \gamma^{t-1} \left(V_t - \sum_{i \in \mathcal{P}_t^{\text{active}}} x_{i,t} - \sum_{i \in \mathcal{P}_t^{\text{active}}} c_{i,t}^{\text{dev}} \right) \right].$$

There is no separate terminal sum. Completion incentives are embedded structurally: the retention amount R_i^{ret} and the final milestone payment R_{i,M_i}^{net} enter through V_t at the period in which completion is certified. Projects that terminate early contribute R_i^{term} to V_t at the termination period, which may be negative, creating a natural deterrent against premature abandonment.

In Level 1, the expectation $\mathbb{E}[\cdot]$ collapses to a deterministic evaluation under $\bar{\eta}_{i,t}$; the stochastic operator is retained here to preserve continuity across all three levels.

The constraints on $\{x_{i,t}\}$ are:

$$x_{i,t} \geq 0, \quad \forall i \in \mathcal{P}, t \in \mathcal{T}, \quad (1)$$

$$\sum_{i \in \mathcal{P}_t^{\text{active}}} x_{i,t} \leq B_t, \quad \forall t \in \mathcal{T}, \quad (2)$$

$$x_{i,t} \geq m_i^0 \cdot \mathbf{1}\{\text{status}_{i,t} = \text{active}\}, \quad \forall i \in \mathcal{P}, t \in \mathcal{T}. \quad (3)$$

Constraint (??) enforces the contractual floor on active projects: allocations must not fall below m_i^0 while a project remains active, as doing so would constitute a client default on contractual obligations. Once $\text{status}_{i,t} \in \{\text{completed}, \text{terminated}\}$, the indicator is zero and the floor constraint is vacuous. The floor therefore serves exclusively as the mechanism that holds the project in a serviceable state while the termination conditions of Section ?? are evaluated; it has no other role in the feasible set.

These primitives — index sets, project parameters including the planned expenditure profile, project and portfolio state variables, transition dynamics, termination conditions, portfolio aggregation, and the shared objective — form the complete foundation from which each formulation level is constructed. Level 1 specializes them under deterministic full foresight; Level 2 reintroduces stochasticity explicitly; Level 3 reformulates the resulting stochastic program as a CMDP.

2.5 Level 1: Deterministic Full-Foresight Program

Level 1 specializes the shared primitives of Section ?? under the assumption of complete information. All budget-to-progress efficiencies $\eta_{i,t}$ are replaced by their known deterministic counterparts $\bar{\eta}_{i,t}$, calibrated from each project’s planned performance profile such

that under nominal execution $\bar{\eta}_{i,t} = 1$. With $\bar{\eta}_{i,t}$ fixed, the progress recursion becomes a deterministic linear map from allocation decisions to physical outcomes, and the active sets $\mathcal{P}_t^{\text{active}}$ are derivable ex ante by tracing the implied completion and termination timing forward through the full horizon. Milestone certification times, advance recovery schedules, retention release dates, and termination settlement payments are consequently all known before any decision is made. The resulting program is a Mixed-Integer Linear Program (MILP) in the budget allocation variables $\{x_{i,t}\}$ and the milestone certification binaries $\{u_{i,j,t}\}$: continuous budget variables appear linearly in every constraint, and the only nonlinearity in the original dynamics — the indicator function governing milestone certification — is captured exactly by the binary formulation introduced below.

Because Level 1 optimizes over the complete planning horizon with perfect knowledge of all outcomes, no causal policy operating under uncertainty can achieve a higher objective value. The Level 1 optimum therefore constitutes a theoretical performance ceiling for the portfolio allocation problem and anchors the empirical comparison in Section ??.

Additional Level 1 Notation

One binary variable and one derived parameter are introduced exclusively for this level.

$u_{i,j,t} \in \{0, 1\}$ Milestone certification indicator: $u_{i,j,t} = 1$ if and only if milestone $j \in \mathcal{M}_i$ of project i has been certified by the end of period t . This binary encodes the state of $\mathcal{M}_i^{\text{cert}}(t)$ in a form compatible with the LP coupling constraints.

$\bar{\eta}_{i,t} > 0$ Deterministic budget-to-progress efficiency for project i at period t , treated as a known parameter at Level 1. Under the nominal plan, $\bar{\eta}_{i,t} = 1$ for all $t \in \mathcal{A}_i^{\text{plan}}$.

$\text{act}_{i,t} \in \{0, 1\}$ Activity indicator derived from the deterministic efficiency trajectory: $\text{act}_{i,t} = 1$ if project i is active at period t , and 0 otherwise. Because $\bar{\eta}_{i,t}$ is known, these indicators are fixed parameters, not decision variables.

All other notation is as defined in Section ?? and Tables ?? and ??.

The Level 1 Program

Objective. The expectation in the shared objective (Section ??) collapses to a deterministic evaluation under $\bar{\eta}_{i,t}$. Defining the per-period net cash-flow expression explicitly:

$$\begin{aligned} \max_{\{x_{i,t}, u_{i,j,t}\}} \quad & \sum_{t=1}^H \gamma^{t-1} \left(\underbrace{\sum_{i \in \mathcal{P}} \sum_{j \in \mathcal{M}_i} R_{i,j}^{\text{net}} (u_{i,j,t} - u_{i,j,t-1}) + \sum_{i \in \mathcal{P}} R_i^{\text{ret}} (u_{i,M_i,t} - u_{i,M_i,t-1}) + \sum_{i \in \mathcal{P}} R_i^{\text{term}} \cdot \mathbf{1}\{t = t_i^{\text{term}}\}}_{V_t} \right. \\ & \left. - \sum_{i \in \mathcal{P}} \text{act}_{i,t} x_{i,t} - \sum_{i \in \mathcal{P}} \text{act}_{i,t} \min\left\{\lambda_i |\delta_{i,t}|, \Lambda_i \cdot CP_i\right\} \right), \end{aligned} \quad (4)$$

where $u_{i,j,0} = 0$ for all i, j and $\delta_{i,t}$ is a linear function of $\{x_{i,\tau}\}_{\tau \leq t}$ through the progress recursion below. The min operator in the penalty term is linearized in constraint (??).

Budget Feasibility and Cash Balance Recursion.

$$B_{t+1} = B_t - \sum_{i \in \mathcal{P}} \text{act}_{i,t} x_{i,t} + V_t, \quad \forall t \in \mathcal{T}, \quad (5)$$

$$\sum_{i \in \mathcal{P}} \text{act}_{i,t} x_{i,t} \leq B_t, \quad \forall t \in \mathcal{T}, \quad (6)$$

$$B_t \geq 0, \quad \forall t \in \mathcal{T}. \quad (7)$$

Progress Recursion and Completion.

$$P_{i,t}^{\text{actual}} = P_{i,t-1}^{\text{actual}} + \bar{\eta}_{i,t} \cdot \frac{x_{i,t}}{\text{BAC}_i}, \quad \forall i \in \mathcal{P}, t \in \mathcal{T}, \quad (8)$$

$$P_{i,t}^{\text{actual}} \leq 1, \quad \forall i \in \mathcal{P}, t \in \mathcal{T}, \quad (9)$$

$$P_{i,0}^{\text{actual}} = 0, \quad \forall i \in \mathcal{P}. \quad (10)$$

Milestone Certification. Let $\mathbf{M} = \max_{i,j} \{\theta_{i,j} \cdot \text{BAC}_i / \bar{\eta}_{i,t}\}$ be a valid big- M constant. Milestone j of project i is certified by period t if and only if cumulative progress has reached or exceeded the required threshold $\theta_{i,j}$:

$$P_{i,t}^{\text{actual}} \geq \theta_{i,j} \cdot u_{i,j,t} - \mathbf{M}(1 - u_{i,j,t}), \quad \forall i \in \mathcal{P}, j \in \mathcal{M}_i, t \in \mathcal{T}, \quad (11)$$

$$P_{i,t}^{\text{actual}} \leq \theta_{i,j} + \mathbf{M} u_{i,j,t}, \quad \forall i \in \mathcal{P}, j \in \mathcal{M}_i, t \in \mathcal{T}, \quad (12)$$

$$u_{i,j,t} \geq u_{i,j,t-1}, \quad \forall i \in \mathcal{P}, j \in \mathcal{M}_i, t \in \mathcal{T}, \quad (13)$$

$$u_{i,j+1,t} \leq u_{i,j,t}, \quad \forall i \in \mathcal{P}, j \in \{1, \dots, M_i - 1\}, t \in \mathcal{T}, \quad (14)$$

$$u_{i,j,0} = 0, \quad \forall i \in \mathcal{P}, j \in \mathcal{M}_i. \quad (15)$$

Cash-Flow Mechanics. Contractual cash inflows at each period are assembled from three deterministic components. The advance payment is received at project initiation:

$$V_t^{\text{adv}} = \sum_{i \in \mathcal{P}} A_i \cdot \mathbf{1}\{t = s_i\}, \quad \forall t \in \mathcal{T}. \quad (16)$$

Milestone net payments (after retention and advance recovery) are triggered at the first certification period:

$$V_t^{\text{ms}} = \sum_{i \in \mathcal{P}} \sum_{j \in \mathcal{M}_i} R_{i,j}^{\text{net}} (u_{i,j,t} - u_{i,j,t-1}), \quad \forall t \in \mathcal{T}, \quad (17)$$

$$\text{where } R_{i,j}^{\text{net}} = \phi_{i,j} \cdot CP_i(1 - \rho_i) - \phi_{i,j} \cdot A_i, \quad \forall i \in \mathcal{P}, j \in \mathcal{M}_i. \quad (18)$$

Retention is released and the final milestone payment made upon completion (certification of $j = M_i$):

$$V_t^{\text{ret}} = \sum_{i \in \mathcal{P}} (R_i^{\text{ret}} + R_{i,M_i}^{\text{net}}) (u_{i,M_i,t} - u_{i,M_i,t-1}), \quad \forall t \in \mathcal{T}. \quad (19)$$

Termination settlement is received at the known termination period t_i^{term} , derivable from the deterministic efficiency trajectory:

$$V_t^{\text{term}} = \sum_{i \in \mathcal{P}} R_i^{\text{term}} \cdot \mathbf{1}\{t = t_i^{\text{term}}\}, \quad \forall t \in \mathcal{T}, \quad (20)$$

$$\text{where } R_i^{\text{term}} = P_{i,t_i^{\text{term}}}^{\text{actual}} \cdot CP_i(1 - \rho_i) - A_i^{\text{outstanding}}, \quad \forall i \in \mathcal{P}^{\text{term}}, \quad (21)$$

$$A_i^{\text{outstanding}} = A_i - \sum_{j \in \mathcal{M}_i} \phi_{i,j} \cdot A_i \cdot u_{i,j,t_i^{\text{term}}}, \quad \forall i \in \mathcal{P}^{\text{term}}. \quad (22)$$

The aggregate period inflow entering the balance recursion (??) is:

$$V_t = V_t^{\text{adv}} + V_t^{\text{ms}} + V_t^{\text{ret}} + V_t^{\text{term}}, \quad \forall t \in \mathcal{T}. \quad (23)$$

Deviation Penalty Linearization. The symmetric running penalty involves an absolute value and a min operator, both of which are linearized by introducing auxiliary variables $d_{i,t}^+, d_{i,t}^- \geq 0$ for the positive and negative parts of the deviation, and $c_{i,t}^{\text{dev}} \geq 0$ for the per-period penalty:

$$\delta_{i,t} = P_{i,t}^{\text{actual}} - P_{i,t}^{\text{plan}}, \quad \forall i \in \mathcal{P}, t \in \mathcal{T}, \quad (24)$$

$$\delta_{i,t} = d_{i,t}^+ - d_{i,t}^-, \quad \forall i \in \mathcal{P}, t \in \mathcal{T}, \quad (25)$$

$$|\delta_{i,t}| = d_{i,t}^+ + d_{i,t}^-, \quad \forall i \in \mathcal{P}, t \in \mathcal{T}, \quad (26)$$

$$c_{i,t}^{\text{dev}} \geq \lambda_i(d_{i,t}^+ + d_{i,t}^-), \quad \forall i \in \mathcal{P}, t \in \mathcal{T}, \quad (27)$$

$$c_{i,t}^{\text{dev}} \leq \Lambda_i \cdot CP_i, \quad \forall i \in \mathcal{P}, t \in \mathcal{T}, \quad (28)$$

$$\Delta_{i,t}^{\text{pen}} = \sum_{\tau=s_i}^t c_{i,\tau}^{\text{dev}}, \quad \forall i \in \mathcal{P}, t \in \mathcal{T}. \quad (29)$$

Termination Conditions. In Level 1, termination timing t_i^{term} is derivable from the known efficiency trajectory. For completeness, the two conditions of Section ?? under deterministic foresight are:

$$\Delta_{i,t_i^{\text{term}}}^{\text{pen}} \geq \kappa_i \cdot \Lambda_i \cdot CP_i, \quad \forall i \in \mathcal{P}^{\text{term}}, \quad (30)$$

$$x_{i,\tilde{t}} \leq m_i^0, \quad \forall \tilde{t} \in \{t_i^{\text{term}} - k + 1, \dots, t_i^{\text{term}}\}, \quad \forall i \in \mathcal{P}^{\text{term}}. \quad (31)$$

Allocation Floor.

$$x_{i,t} \geq m_i^0 \cdot \text{act}_{i,t}, \quad \forall i \in \mathcal{P}, t \in \mathcal{T}. \quad (32)$$

Variable Domains.

$$x_{i,t} \geq 0, \quad \forall i \in \mathcal{P}, t \in \mathcal{T}, \quad (33)$$

$$u_{i,j,t} \in \{0, 1\}, \quad \forall i \in \mathcal{P}, j \in \mathcal{M}_i, t \in \mathcal{T}, \quad (34)$$

$$d_{i,t}^+, d_{i,t}^-, c_{i,t}^{\text{dev}}, P_{i,t}^{\text{actual}}, B_t \geq 0, \quad \forall i \in \mathcal{P}, t \in \mathcal{T}. \quad (35)$$

Economic Interpretation

The constraint groups are interpreted in turn.

Budget feasibility and cash balance recursion (??)–(??). Constraint (??) propagates the portfolio cash balance forward by one period: it deducts total active-project expenditure from the opening balance and adds all contractual inflows received in period t . Constraint (??) enforces the hard spending limit: the total allocated budget in any period cannot exceed the cash available at the start of that period. Constraint (??) prohibits borrowing, consistent with the fixed-capital governance structure described in Section ???. Together, these three constraints encode the portfolio manager’s primary operational constraint: all cash out must be covered by cash in, period by period.

Progress recursion and completion (??)–(??). Constraint (??) is the deterministic collapse of the budget-to-progress conversion from Section ???: a budget allocation $x_{i,t}$ generates a proportional progress increment scaled by the known efficiency $\bar{\eta}_{i,t}$ and normalized by BAC_i . Constraint (??) enforces the natural upper bound that no project can exceed full completion; this prevents the optimizer from over-allocating budget to drive nominal progress beyond 100%. Constraint (??) initializes all projects at zero cumulative progress at the start of the horizon.

Milestone certification (??)–(??). These constraints implement the certification logic of $\mathcal{M}_i^{\text{cert}}(t)$ in binary form. Constraints (??) and (??) together enforce the equivalence: $u_{i,j,t} = 1$ if and only if $P_{i,t}^{\text{actual}} \geq \theta_{i,j}$. Constraint (??) imposes temporal monotonicity — once a milestone is certified it cannot be decertified, which is a contractual requirement under FIDIC forms ?. Constraint (??) enforces that milestones can only be certified in sequence, consistent with their ordered threshold structure $\theta_{i,1} < \dots < \theta_{i,M_i}$. Constraint (??) initializes all milestones as uncertified.

Cash-flow mechanics (??)–(??). These constraints translate the contractual payment structure from Section ?? into deterministic linear expressions. Constraint (??) delivers the advance payment as a fixed inflow at the planned start date s_i ; it enters the cash balance recursion immediately and reduces the net financing requirement of the project. Constraints (??)–(??) compute milestone inflows as the difference in certification binaries ($u_{i,j,t} - u_{i,j,t-1}$), which equals one in exactly the period the milestone is first certified and zero thereafter. This formulation linearizes the milestone trigger without introducing additional binary variables beyond $u_{i,j,t}$. Constraint (??) similarly delivers the retention release and final milestone payment in the period the last milestone is certified. Constraints (??)–(??) model termination settlement as a known inflow at t_i^{term} : the settlement value equals the progress fraction of the contract price, less any outstanding advance balance, both of which are deterministic functions of the known efficiency trajectory.

Deviation penalty linearization (??)–(??). Constraints (??)–(??) decompose the signed progress deviation $\delta_{i,t}$ into non-negative positive and negative parts, enabling the absolute value to be handled linearly. Constraint (??) lower-bounds the per-period penalty variable $c_{i,t}^{\text{dev}}$ by the weighted absolute deviation; since the objective minimizes total penalty (equivalently, maximizes the negated penalty), the optimizer drives $c_{i,t}^{\text{dev}}$ to this bound, recovering the exact symmetric penalty without any nonlinearity. Constraint (??)

imposes the contractual ceiling $\Lambda_i \cdot CP_i$, consistent with the cap defined in Table ?? . Constraint (??) accumulates the per-period penalties into the running total $\Delta_{i,t}^{\text{pen}}$, which feeds the termination condition.

Termination conditions (??)–(??). These constraints express the two-condition termination gate of Section ?? under deterministic foresight. Because $\bar{\eta}_{i,t}$ is known, the period t_i^{term} at which both conditions are first simultaneously satisfied is a fixed parameter derivable from the efficiency trajectory before solving the MILP. The MILP therefore treats these constraints as verifiable parameter conditions rather than active optimization constraints; they are included for formal completeness and to preserve structural continuity with the Level 2 and Level 3 formulations, where termination timing is endogenous and these constraints are binding.

Allocation floor (??). This constraint enforces the minimum budget required to hold each active project in a contractually serviceable state. As established in Section ??, m_i^0 has no performance-recovery role; it is the floor allocation at which the termination conditions are evaluated without constituting a client default. The $\text{act}_{i,t}$ indicator, fixed as a parameter at Level 1, ensures the floor applies only while the project is active: it is vacuous once $\text{status}_{i,t} \in \{\text{completed}, \text{terminated}\}$.

The Upper Bound Argument

Let Z_{L1}^* denote the optimal objective value of the Level 1 program. The claim is that for any causal allocation policy π operating under uncertainty, the expected objective value Z^π satisfies $Z^\pi \leq Z_{\text{L1}}^*$.

The argument is direct. The feasible set of the Level 1 MILP is at least as large as the feasible set available to any causal policy: a causal policy can only condition allocations on information revealed up to the current period, whereas the Level 1 program conditions on all future information simultaneously. Any feasible allocation sequence generated by a causal policy under any realization of the efficiency process is also feasible in Level 1 — since Level 1 is solved for the realized trajectory, not in expectation. Taking the expectation over realizations and the supremum over policies does not change this ordering: $\sup_{\pi} \mathbb{E}[Z^\pi] \leq Z_{\text{L1}}^*$. The Level 1 optimum therefore constitutes a strict benchmark that no implementable policy can attain in expectation, and against which the PPO agent, the rule-based heuristic, and the two-stage stochastic program are all evaluated in Section ?? . The gap between Z_{L1}^* and the achieved objective of each policy quantifies the cost of operating under uncertainty relative to an oracle with perfect foresight.